



Summer 2026

Computer Science & Engineering Department

CSE412: Software Engineering (Section 01)

TEAMUP — Academic Project Team Formation & Collaboration Platform

Software Requirements Specification (SRS) Report

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Chapter 1: Introduction

1.1: Project Overview (User Requirements Story)

TEAMUP is a web-based academic project collaboration platform designed to support undergraduate students and faculty members at East West University. It serves three primary user roles: Students, Supervisors, and Administrators, each accessing the system through a secure login that leads to a personalized dashboard. The purpose of the platform is to centralize the entire academic project lifecycle, from finding teammates and securing a supervisor, to managing tasks, sharing files, and showcasing completed work, so that students no longer have to rely on scattered tools like WhatsApp, Facebook, and email.

A student registers using their student ID and a platform-specific password. After logging in, students who have not yet joined a group are welcomed by a discovery feed resembling an academic social network, showing other students looking for groups, open recruitment posts, and supervisor profiles. Each student builds a profile with their skill set, CGPA, completed credits, domain interest, target semester, and additional credentials like published papers or RA/TA experience, along with their current group status.

Supervisors similarly build professional profiles listing their department, research interests, domains they supervise, and whether they are accepting new groups. Students can browse and filter supervisor profiles before sending a formal supervision request. Supervisors accept or decline with an optional note and can take a maximum of two to three groups per semester.

Students search for teammates by skills, domain, preferred supervisor, or target semester, communicate via direct messaging, and either start a group or join one by invitation. As the group forms, a skill gap visualiser shows which roles are covered and which are missing. A group chat is created automatically once the team is finalised. If a member leaves mid-project, the team can repost a vacancy with full project context for interested applicants.

Once a supervisor accepts the team's request, they are added to the collaboration workspace. The workspace supports a structured task board with type-tagged tasks, a proposal workflow from draft to approval, milestone markers, a progress bar, and organised file sharing for code, reports, presentations, and datasets. Supervisors have read-only access to all content and can post announcements and approve milestones.

At project completion, teams can publish their work to a public showcase with descriptions, tech stack tags, screenshots, and member profiles. The public showcase is also accessible to visitors outside the platform, giving completed projects broader visibility. A full progress summary can also be exported as a PDF. Administrators manage account verification, content moderation, and university data in the background.

1.2 Purpose and Scope

TEAMUP centralizes the academic project lifecycle for undergraduate students and faculty at East West University, replacing informal communication tools with a structured, role-based web platform covering team formation, supervisor coordination, project management, and outcome documentation.

The platform supports student and supervisor profile discovery, group formation with skill gap analysis, formal supervisor assignment, a collaboration workspace with task and file management, and end-of-project showcase and export features. It is scoped for undergraduate course-based or capstone projects within a single university environment and does not cover postgraduate supervision or integration with existing university management systems in this version.

1.3 Stakeholders

Primary stakeholders are students, who use the platform to find teams, manage projects, and document project outcomes, and supervisors, who oversee teams, review proposals, provide guidance and monitor project progress throughout the project lifecycle.

Secondary stakeholders include university administrators, who are responsible for account management, user verification, maintaining platform data, and ensuring the smooth operation of the system. Department Coordinators and Evaluators can access project records and history for assessment purposes, while prospective employers may view the public project showcase as evidence of a student's academic skills and experience.

Chapter 2: Requirements Engineering Process

2.1: Stakeholder Needs & Analysis

Primary Stakeholders

Primary stakeholders are those who directly interact with and are most affected by the platform on a regular basis.

Students:

- Main users of the platform who form teams, manage projects, communicate with supervisors, and document outcomes.
- Need a simple way to find compatible teammates and supervisors, manage project tasks, share files, and build a portfolio of completed work.

Supervisors / Faculty:

- Oversee student teams, review and approve project proposals, monitor task progress, and provide feedback.
- Need clear visibility into team activity and contributions, a structured proposal review process, and an easy way to manage their group load per semester.

Secondary Stakeholders

Secondary stakeholders are those who are indirectly affected by the platform or have an interest in its outcomes without being regular users.

Administrators:

- Manage the platform behind the scenes by verifying accounts, moderating content, and maintaining university and semester data.
- Need reliable tools to manage user accounts, handle disputes, and keep platform data accurate across semesters.

Department Coordinators and Evaluators:

- May reference project records, team formation history, and the public showcase for academic evaluation and quality assessment.
- Need access to accurate and organised project data and outcomes for assessment purposes.

Prospective Employers and Recruiters:

- May view the public project showcase as a reference for a student's academic experience and technical skills.
- Need a credible, publicly accessible record of student project work and team contributions.

Methods Used for Requirement Elicitation

Requirements were gathered using a hybrid approach combining structured interviews, a student survey and some informal faculty feedback.

- **Interviews:** Conducted with classmates and peers (undergraduate CSE students at East West University) who regularly participate in semester project formation. Discussions focused on key challenges in:
 - Finding suitable teammates
 - Approaching supervisors
 - Managing tasks
 - Submitting academic deliverables
- **Survey:** A structured survey was conducted on 30 university students (n=30):
 - 87% reported moderate to high difficulty in finding suitable teammates (Figure 2.1)
 - 73% identified skill mismatch, uneven contribution, and scattered tools as major issues (Figure 2.2)
 - Top requested features included:
 - Skill-based teammate matching
 - Skill gap visualizer
 - Shared task management board (Figure 2.3)
- **Supervisor input:** Informal feedback highlighted limited visibility into team dynamics as a key gap in current workflows.

Overall, these findings directly shaped the platform's core feature set and requirement priorities.

How difficult is it to find suitable teammates for your project courses, capstones, or research?

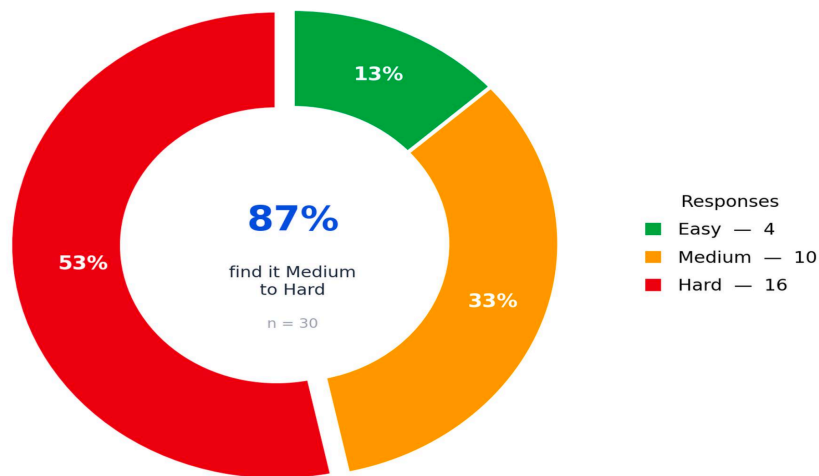


Figure 2.1: Survey Results for difficulty in finding groupmates

Which problems have you faced during group projects?

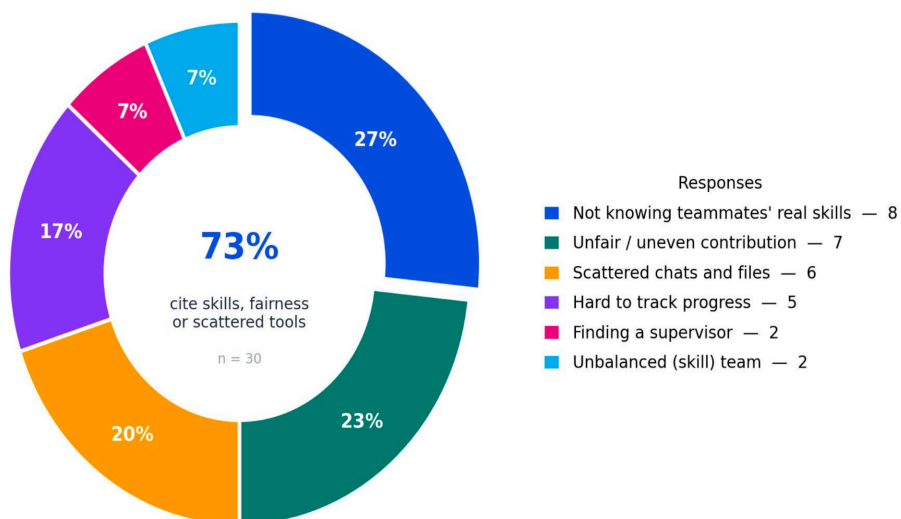


Figure 2.2: Survey Results for problems faced in group projects

Which features would you want most in such a platform?

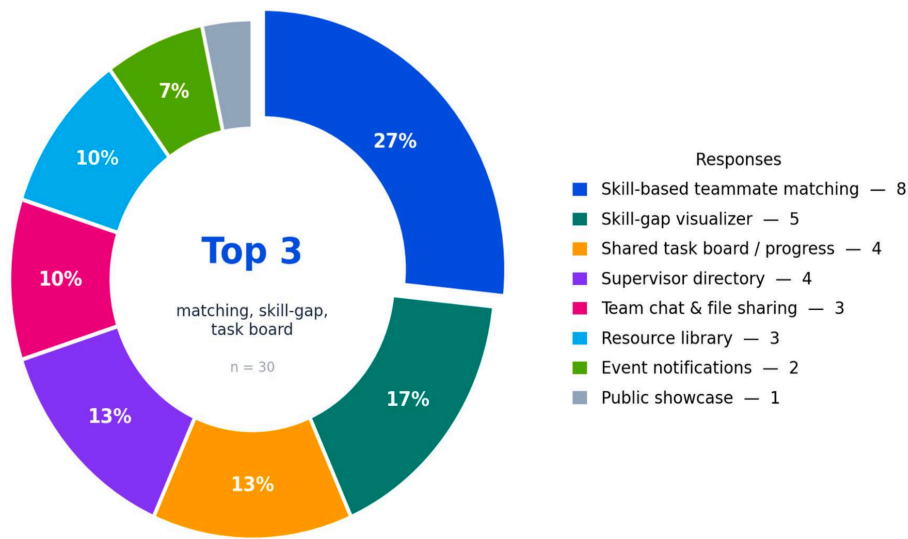


Figure 2.3: Survey Results for expected features in the platform

2.2: List of Requirements

Functional Requirements (FRs)

FR1: The system must allow students, supervisors, and administrators to register and log in using verified university credentials and role-specific dashboards.

FR2: Students must be able to build academic profiles with skills, CGPA, domain interests, and credentials, while supervisors must be able to list research interests, domain preferences, and availability status.

FR3: Students must be able to discover and search other student and supervisor profiles, post recruitment announcements, send direct messages, form groups, and manage team invitations.

FR4: Teams must be able to send a formal supervision request to a supervisor, who can accept or decline, with the supervisor being added to the team workspace upon acceptance.

FR5: The system must provide a collaboration workspace for every team containing a type-tagged task board, proposal workflow tracker, milestone markers, progress bar, and organised file sharing.

FR6: At project completion, teams must be able to publish their work to a public showcase and export a project progress summary as a PDF.

Non-Functional Requirements (NFRs)

NFR1: Security - All user data must be protected through secure authentication, encrypted data transmission, and role-based access control ensuring each user only accesses content relevant to their role.

NFR2: Performance - The platform must load pages within 3 seconds under normal usage and reflect task and status updates promptly without requiring a manual page refresh.

NFR3: Usability - The interface must be clean, intuitive, and accessible on both desktop and mobile browsers without requiring any prior training for new users.

NFR4: Scalability - The system must be capable of supporting all active users of a university simultaneously without performance degradation.

NFR5: Availability - The platform must maintain reliable uptime throughout the academic semester, particularly around project deadlines and submission periods.

NFR6: Data Integrity - All proposal revision histories, task logs, milestone records, and contribution data must be permanently stored and protected from unauthorized modification or deletion.

Extraordinary Requirements (Wow Factors)

EX1: Skill Gap Visualiser - When forming a team, the platform displays a live visual chart showing which required roles and skills are already covered by current members and which are still missing, updating automatically as new members join.

EX2: Team Compatibility Summary - When a team is formed, the platform automatically generates a compatibility report based on the combined skill profiles of all members, highlighting skill overlaps, gaps, and role distribution in one snapshot. Helps teams make informed decisions before finalising their group.

EX3: Public Project Showcase - Completed projects can be optionally published to a permanent public gallery with descriptions, tech stack tags, screenshots, and linked member profiles, giving every student a credible portfolio entry that persists beyond the semester.

2.3: House of Quality (QFD Integration)

Customer Requirements (CRs)

- **CR1:** Students need a simple way to find and connect with suitable teammates
- **CR2:** Students need to discover and approach supervisors based on availability and domain
- **CR3:** Teams need an organised workspace to manage tasks and track project progress
- **CR4:** Users need secure and role-appropriate access to platform features
- **CR5:** Teams need a structured way to submit and track project proposals
- **CR6:** Students need a permanent record of their completed academic project work
- **CR7:** Teams need visibility into skill coverage when forming a group

Engineering/Technical Requirements (TRs)

- **TR1:** Implement role-based authentication and verified registration system
- **TR2:** Build searchable profile directory with filters for skills, domain, and availability
- **TR3:** Develop a structured supervision request and acceptance workflow
- **TR4:** Implement a task board with type-tagged tasks, priority levels, and status tracking
- **TR5:** Implement a proposal tracker with status pipeline and supervisor review workflow
- **TR6:** Deploy on a scalable cloud-based backend with real-time data updates
- **TR7:** Build a public showcase page with PDF export functionality
- **TR8:** Develop skill gap visualiser and team compatibility summary using profile data

QFD Matrix (House of Quality):

Table 2.1: QFD Matrix

Customer Requirements (CRs)	Technical Requirements (TRs)	Relationship
CR1: Students need a simple way to find and connect with suitable teammates	TR2: Build searchable profile directory with filters for skills, domain, and availability	Strong
CR1: Students need a simple way to find and connect with suitable teammates	TR8: Develop skill gap visualiser and team compatibility summary using profile data	Strong
CR2: Students need to discover and approach supervisors based on availability and domain	TR2: Build searchable profile directory with filters for skills, domain, and availability	Strong
CR2: Students need to discover and approach supervisors based on availability and domain	TR3: Develop a structured supervision request and acceptance workflow	Strong
CR3: Teams need an organised workspace to manage tasks and track project progress	TR4: Implement a task board with type-tagged tasks, priority levels, and status tracking	Strong
CR3: Teams need an organised workspace to manage tasks and track project progress	TR6: Deploy on a scalable cloud-based backend with real-time data updates	Medium
CR4: Users need secure and role-appropriate access to platform features	TR1: Implement role-based authentication and verified registration system	Strong

CR4: Users need secure and role-appropriate access to platform features	TR6: Deploy on a scalable cloud-based backend with real-time data update	Medium
CR5: Teams need a structured way to submit and track project proposals	TR5: Implement a proposal tracker with status pipeline and supervisor review workflow	Strong
CR5: Teams need a structured way to submit and track project proposals	TR4: Implement a task board with type-tagged tasks, priority levels, and status tracking	Weak
CR6: Students need a permanent record of their completed academic project work	TR7: Build a public showcase page with PDF export functionality	Strong
CR7: Teams need visibility into skill coverage when forming a group	TR8: Develop skill gap visualiser and team compatibility summary using profile data	Strong

Chapter 3: Requirements Modeling

3.1: Use case Diagrams

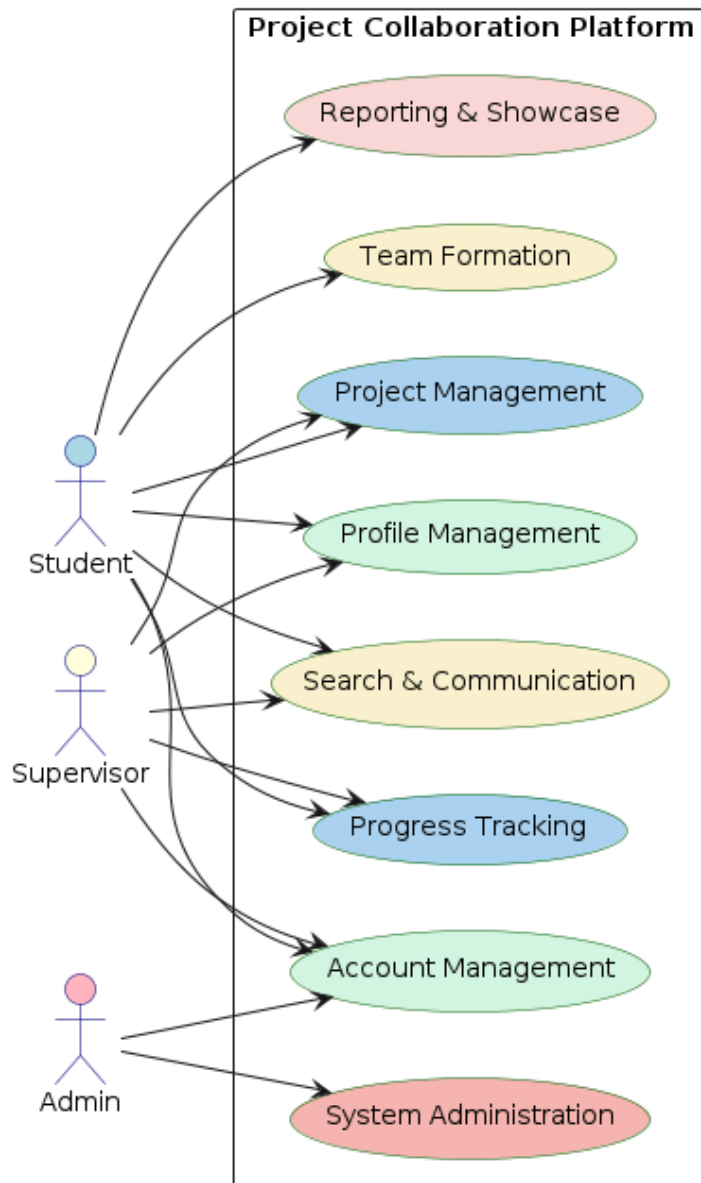


Figure 3.1.1: Level 1 Use Case Diagram - TEAMUP System Overview

3.2: Activity Diagrams

TEAMUP Activity Diagram: Proposal, Task, and Progress Management

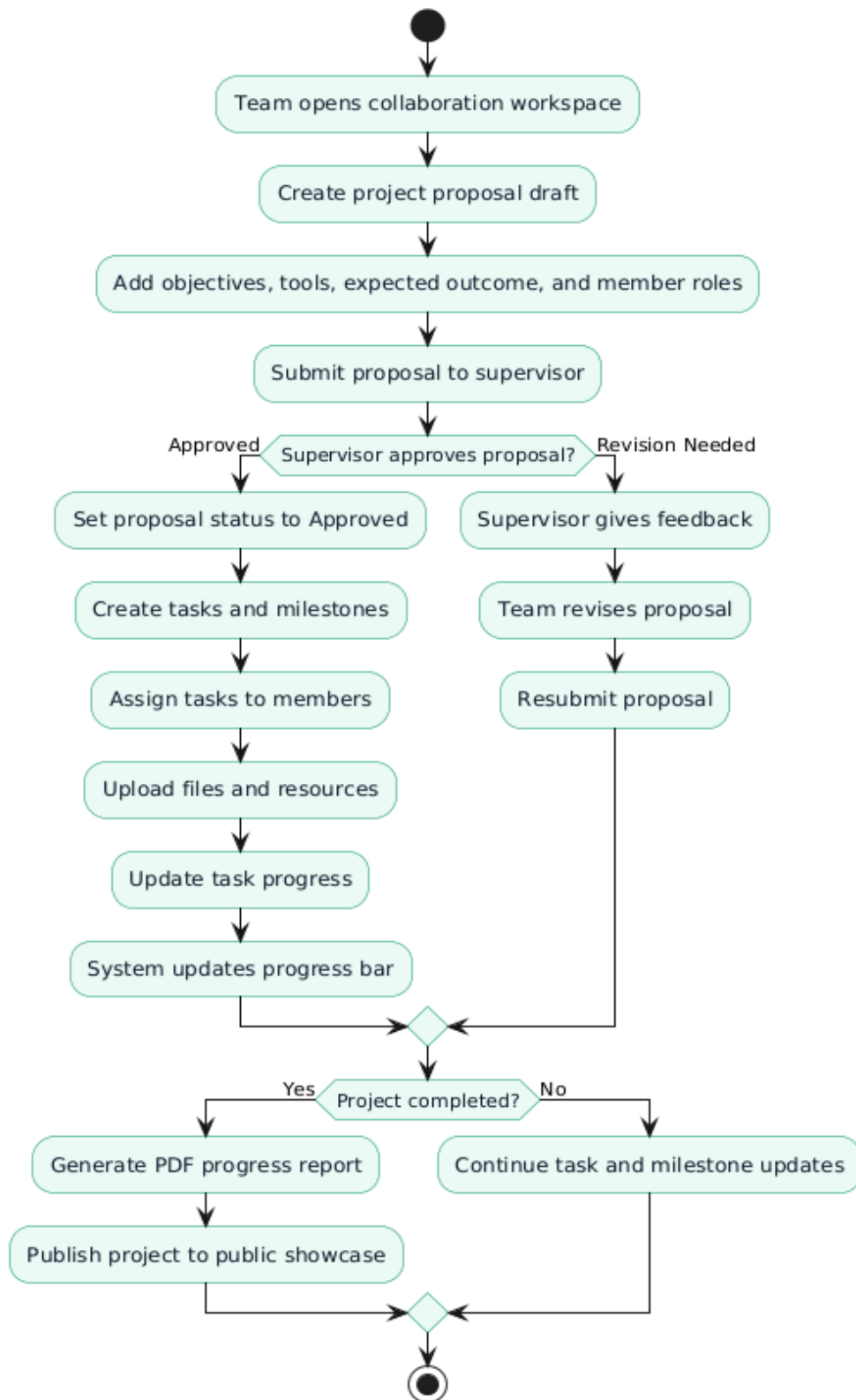


Figure 3.2.1: Activity Diagram of Proposal and Task Management

3.3: Sequence Diagrams

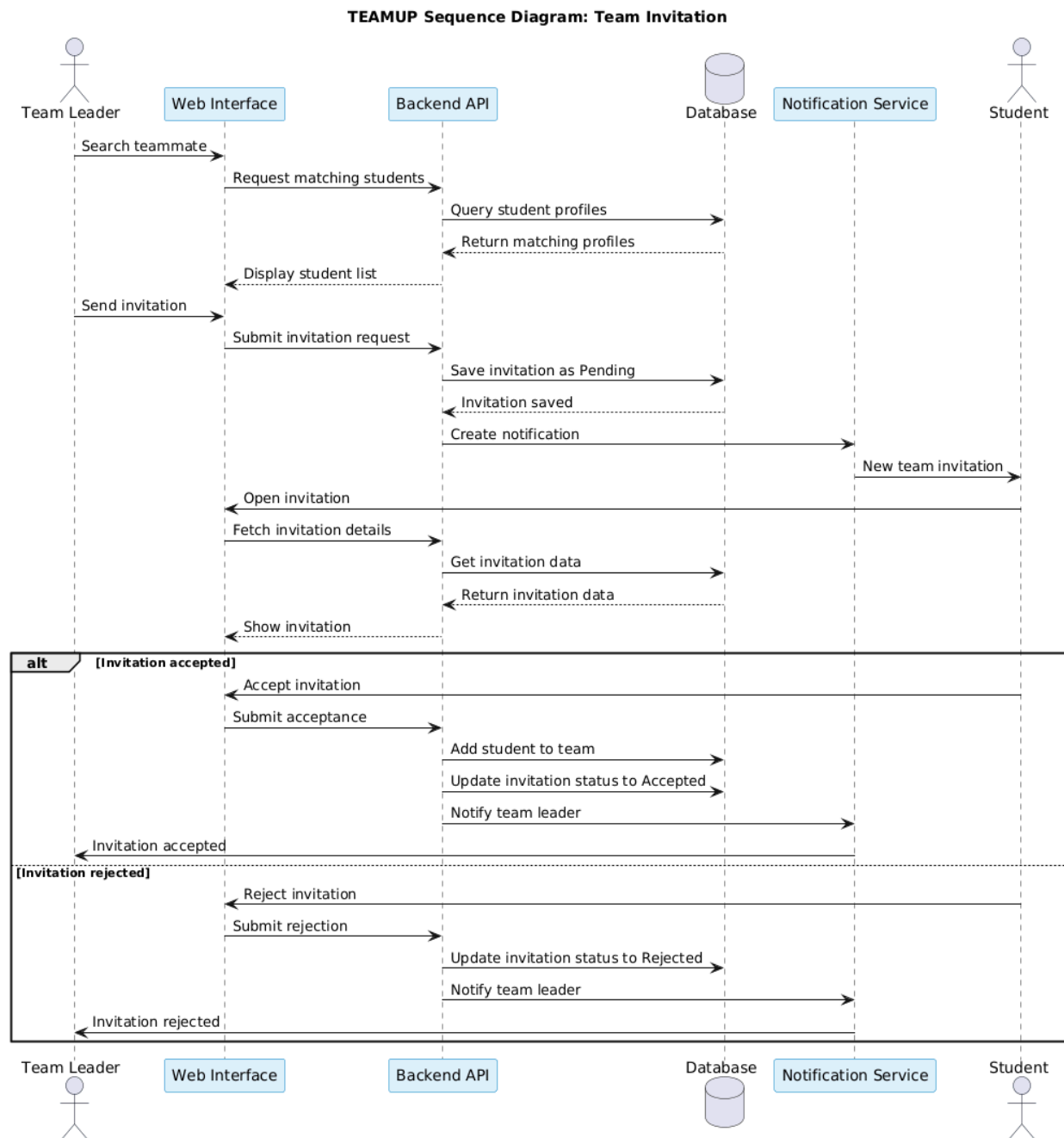


Figure 3.3.1: Sequence Diagram of Team Invitation

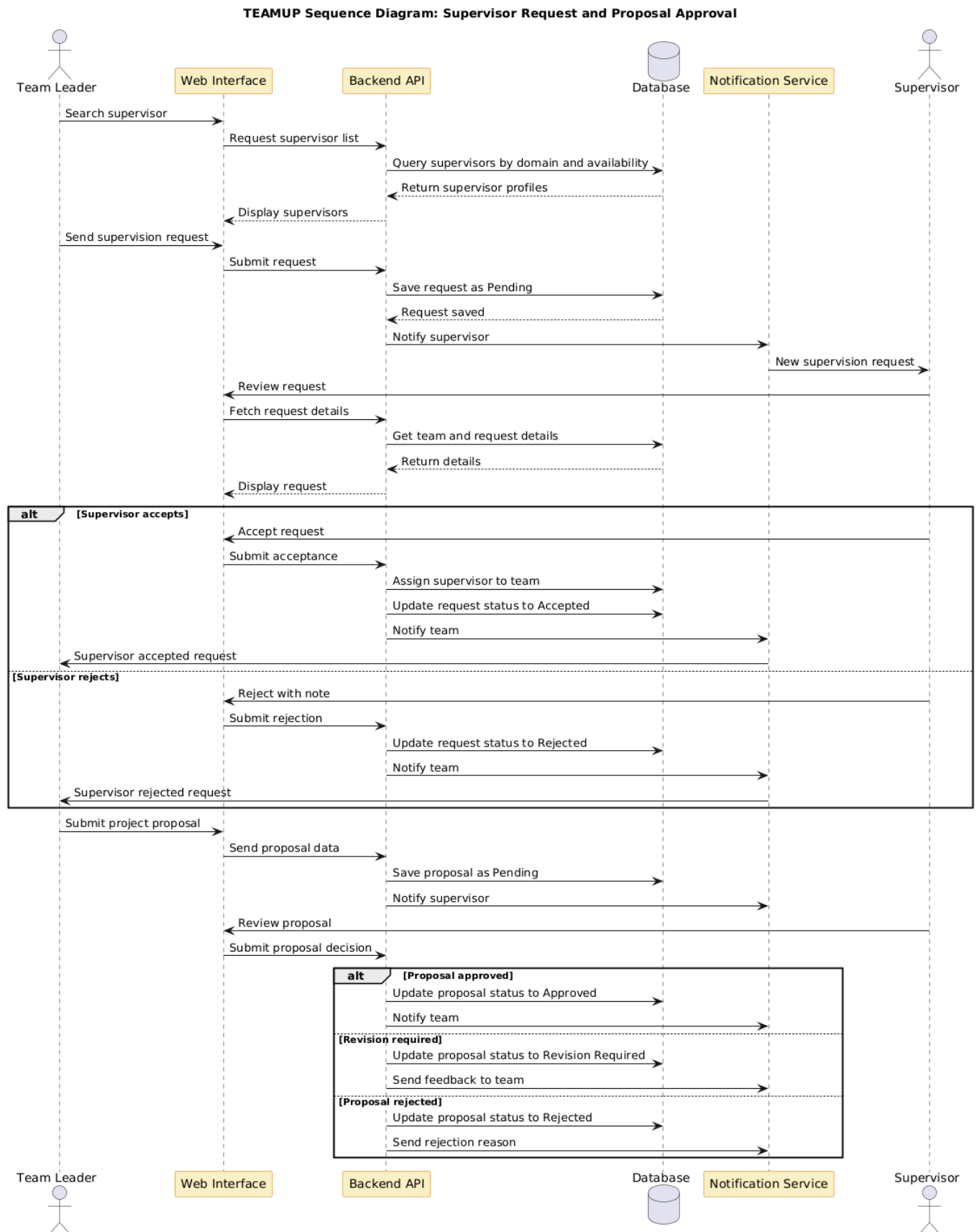


Figure 3.3.2: Sequence Diagram of Supervisor Request and Proposal Approval

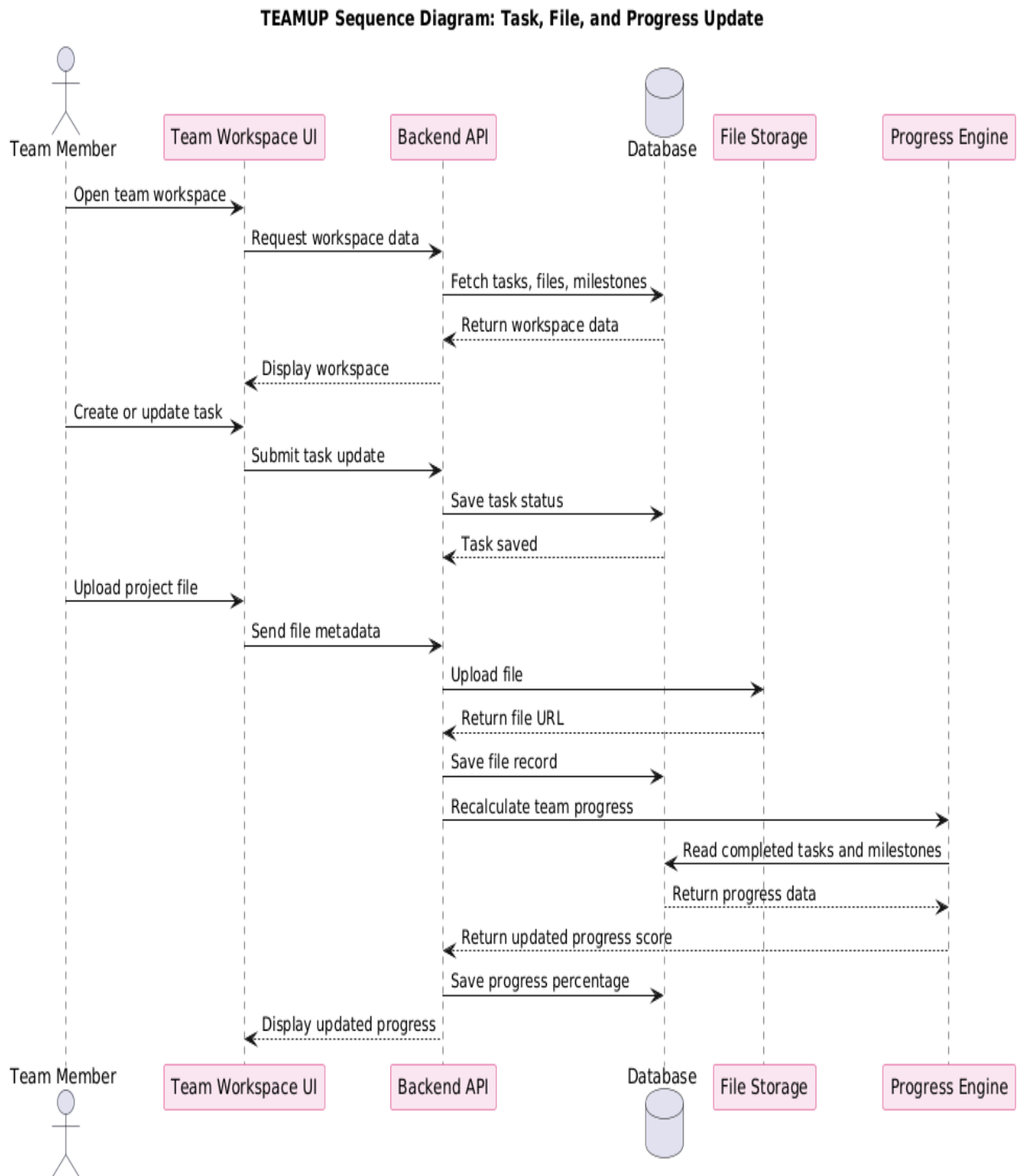


Figure 3.3.3: Sequence Diagram of Task, File and Progress update

3.4: Prototyping using wireframes or UI sketches

Figure 3.4.1 Login and Registration Page

· TEAMUP - Login / Registration

Email:

Password:

Role:

Figure 3.4.2 Student Dashboard

· TEAMUP - Student Dashboard

Welcome, Student

Profile Completion: 85%

Current Team Status: Not Joined / Joined

· Recent Notifications

- Team invitation received
- Supervisor request pending
- Proposal feedback available

Figure 3.4.3 Team Formation Interface

· TEAMUP - Team Formation Interface

Search Skill:

Python / React / ML

Domain:

Data Science

Semester:

Summer 2026

Search

· Recommended Teammates

Name	Skills	Compatibility	Action
Rahim	Python, SQL	92%	Invite
Karim	ML, Power BI	87%	Invite
Nadia	UI, React	84%	Invite

· Skill Gap Visualizer

Covered: Backend, Database

Missing: Frontend, Documentation

Figure 3.4.4 Supervisor Matching Interface

TEAMUP - Supervisor Matching Interface

Project Domain:

Machine Learning

Availability:

Accepting Groups

Find Supervisor

Recommended Supervisors:

Name	Research Area	Action
Dr. Ahmed	ML, Data Mining	Request
Dr. Rahman	AI, NLP	Request
Dr. Karim	Software Eng.	Request

Figure 3.4.5 Team Collaboration Workspace

· TEAMUP - Team Collaboration Workspace

Team Name:

Team Alpha

Supervisor:

Dr. Ahmed

Proposal Status:

Approved

Progress:

75%

Submit Proposal

Upload File

Export PDF

· Task Board

Task

Assigned To

Status

Literature Review

Rahim

Completed

UI Design

Nadia

In Progress

Backend API

Karim

To Do

Report Writing

Tamim

Review

· Shared Files

- proposal.docx

- dataset.csv

- presentation.pptx

Figure 3.4.6 Public Project Showcase

· TEAMUP - Public Project Showcase

Project Title:

Smart Crop Analytics System

Team Name:

Team Alpha

Supervisor:

Dr. Ahmed

Tech Stack:

React, Node.js, MongoDB, ML

Description:

This project predicts crop yield using data analytics and machine learning.

View Project Details

Download Report

View Team Members

· Team Members

Name

Role

Tamim

Data Analyst

Rahim

Backend Developer

Nadia

UI Designer

Karim

ML Developer

Chapter 4: Constraints and Assumptions

4.1: Constraints

The development of TEAMUP is subject to the following constraints:

1. The system will initially be developed as a web-based platform and may not include a dedicated mobile application in the first release.
2. The platform requires a stable internet connection for accessing collaboration features, file sharing, and project management tools.
3. Each student can participate in only one active project team within a specific course at a time.
4. Supervisor assignment is limited by the supervisor's available supervision capacity.
5. File upload size may be restricted to ensure efficient storage and system performance.
6. The project will be developed using available academic resources, development tools, and hosting services.
7. The first version of the system will be designed primarily for use within a single university environment.

4.2: Assumptions

The following assumptions are considered during system design and development:

1. Students will provide accurate academic profiles, skills, and project interests.
2. Supervisors will regularly update their research interests and availability information.
3. Users possess basic knowledge of web applications and online collaboration tools.
4. All users will have valid institutional email addresses for registration and verification.
5. Internet connectivity will be available whenever users access the platform.
6. University authorities will support the use of the system for academic project management.
7. Users will follow ethical guidelines and maintain professional communication within the platform.

Chapter 5: Validation and Testing Plan

To ensure the quality, reliability, and functionality of TEAMUP, various testing techniques will be applied throughout the development lifecycle.

5.1: Unit Testing

Individual system components such as authentication, profile management, team formation, supervisor assignment, and file sharing modules will be tested independently.

5.2: Integration Testing

The interaction between different modules will be tested to verify smooth communication among frontend, backend, and database components.

5.3: Functional Testing

All functional requirements will be validated to ensure the system performs according to specified user requirements.

Examples include:

- User registration and login
- Team creation
- Supervisor request submission
- Proposal approval workflow
- Task management
- Project showcase publication

5.4: Usability Testing

Representative students and supervisors will evaluate the platform to assess:

- Ease of navigation
- User interface clarity
- Accessibility
- Overall user experience

5.5: Performance Testing

The system will be evaluated under expected user loads to ensure:

- ❖ Fast page response times
- ❖ Efficient database operations
- ❖ Stable file upload performance

5.6: Security Testing

Security testing will verify:

- User authentication mechanisms
- Authorization and access control
- Data protection
- Secure file handling

5.7 User Acceptance Testing (UAT)

Final testing will be conducted with selected students and supervisors to confirm that the platform satisfies stakeholder expectations and project objectives.

Chapter 6: Future Enhancements

The following enhancements may be incorporated in future versions of TEAMUP:

1. Mobile Application Development for Android and iOS platforms.
2. AI-Based Teammate Recommendation Engine using machine learning techniques.
3. Advanced Supervisor Recommendation System based on research interests and project history.
4. GitHub Integration for software project collaboration and repository management.
5. Calendar and Deadline Management System with automated reminders.
6. Peer Evaluation and Contribution Assessment Module.
7. Real-Time Video Meeting Integration.
8. Advanced Project Analytics Dashboard.
9. Multi-University Support for inter-university project collaboration.
10. Internship and Research Opportunity Recommendation System.
11. Integration with Learning Management Systems (LMS) such as Moodle and Google Classroom.

Chapter 7: Conclusion

TEAMUP is a comprehensive academic project team formation, supervisor matching, and collaboration platform designed to address common challenges faced by students and faculty members during project-based learning activities.

The platform provides a centralized environment for teammate discovery, supervisor assignment, project collaboration, task management, proposal review, progress monitoring, and project showcasing. By integrating these functionalities into a single system, TEAMUP improves communication, enhances project transparency, and promotes effective academic collaboration.

The proposed system satisfies the identified stakeholder needs, functional requirements, non-functional requirements, and quality objectives defined in this Software Requirements Specification (SRS). Furthermore, the inclusion of innovative features such as skill gap visualization, team compatibility analysis and project showcase functionality enhances the overall value of the platform.

TEAMUP has the potential to significantly improve the academic project management process while providing students with opportunities to build professional portfolios and strengthen collaborative learning experiences.